

MANISH KUMAR

AI Research Engineer | Agentic AI • Cognitive Architectures

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🔗 manish-portfolio-one-gamma.vercel.app

Profile Summary

AI Research Engineer interested in Cognitive AI, Autonomous Agents, Reinforcement Learning, and Graph-Based Memory Systems. Building research-oriented AI systems combining semantic memory, planning, large language models, and adaptive reasoning to explore interpretable cognitive architectures.

Education

Minor in Artificial Intelligence & Data Science, 05/2025 – 05/2026
Indian Institute of Technology (IIT) Mandi
CGPA: 7.5/10

Introduction to Web Development with HTML, CSS, JavaScript, IBM 04/2023 – 03/2025

Skills

- **Languages:** Python, JavaScript, SQL
- **AI/ML:** PyTorch, TensorFlow, Reinforcement Learning, RAG, Embeddings, LangGraph, Ollama APIs, Anthropic APIs
- **Frameworks/Libraries:** FastAPI, Three.js, NumPy, Pandas
- **Data & Infrastructure:** SQLite, ChromaDB (vector store), Graph-Based Memory Structures
- **Tools:** Git, GitHub, VS Code
- **Soft Skills:** Analytical Problem Solving, Research Mindset, Systems Thinking, Self-Learning, Adaptability, Technical Communication

Projects

MiniFlyWire - Brain-inspired cognitive architecture 🔄 05/2026 – Present
Tech Stack: JavaScript • Three.js • WebGL • HTML5 • CSS3 • Git • GitHub

- Designed a brain-inspired cognitive architecture that navigates a semantic concept graph comprising 20 interconnected concepts using goal-directed attention, curiosity-driven exploration, and biologically inspired executive control.
- Built a Q-learning-based adaptive decision system integrating rewards, penalties, episodic memory, attention mechanisms, and future planning to continuously improve navigation strategies.
- Engineered a modular memory architecture integrating working memory, episodic memory, semantic memory, long-term memory consolidation, and embedding-based knowledge representation.
- Created an interactive Three.js-based 3D cognitive observatory that visualizes neural activation, memory formation, attention dynamics, and real-time decision-making processes.

Noetica - Adaptive Learning Intelligence Engine 🔄 06/2026 – Present
Tech Stack: Python • Ollama • Qwen3:4B • REST API • Pytest • Git • GitHub • FastAPI • SQLite • ChromaDB • NumPy

- Architected the foundational cognitive architecture for an adaptive learning intelligence engine using modular AI agent design principles.
- Built a provider-agnostic LLM abstraction layer integrating Ollama and Qwen3:4B with structured response handling and reasoning separation.
- Designed a modular agent architecture separating reasoning, memory, planning, and evaluation components for future multi-agent collaboration.
- Implemented deterministic response sanitization, token management, and unit-tested processing pipelines to improve LLM reliability.